

Replication

1. Берем тестовый датасет YouTube

```
CREATE TABLE youtube
(
    `id` String,
    `fetch_date` DateTime,
    `upload_date_str` String,
    `upload_date` Date,
    `title` String,
    `uploader_id` String,
    `uploader` String,
    `uploader_sub_count` Int64,
    `is_age_limit` Bool,
    `view_count` Int64,
    `like_count` Int64,
    `dislike_count` Int64,
    `is_crawlable` Bool,
    `has_subtitles` Bool,
    `is_ads_enabled` Bool,
    `is_comments_enabled` Bool,
    `description` String,
    `rich_metadata` Array(Tuple(call String, content String, subtitle String, title String, url String)),
    `super_titles` Array(Tuple(text String, url String)),
    `uploader_badges` String,
    `video_badges` String
)
ENGINE = MergeTree
ORDER BY (uploader, upload_date);
```

2_Загружаем данные

```
INSERT INTO task_15.youtube
SELECT id,
    parseDateTimeBestEffortUS0rZero(toString(fetch_date)) AS fetch_date,
    upload_date AS upload_date_str,
    toDate(parseDateTimeBestEffortUS0rZero(upload_date::String)) AS upload_date,
    ifNull(title, '') AS title,
    uploader_id,
    ifNull(uploader, '') AS uploader,
    uploader_sub_count,
    is_age_limit,
    view_count,
    like_count,
    dislike_count,
    is_crawlable,
    has_subtitles,
    is_ads_enabled,
    is_comments_enabled,
    ifNull(description, '') AS description,
    rich_metadata,
    super_titles,
    ifNull(uploader_badges, '') AS uploader_badges,
    ifNull(video_badges, '') AS video_badges
FROM s3('https://clickhouse-public-datasets.s3.amazonaws.com/youtube/original/files/*.zst',
    'JSONLines')
LIMIT 10000
SETTINGS input_format_null_as_default = 1;
```

3_ Конвертируем и добавляем 2 реплики

```
CREATE DATABASE replicated ON CLUSTER replicated_cluster;

CREATE TABLE IF NOT EXISTS replicated.youtube ON CLUSTER replicated_cluster
(
    `id` String,
    `fetch_date` DateTime,
    `upload_date_str` String,
    `upload_date` Date,
    `title` String,
    `uploader_id` String,
    `uploader` String,
    `uploader_sub_count` Int64,
    `is_age_limit` Bool,
    `view_count` Int64,
    `like_count` Int64,
    `dislike_count` Int64,
    `is_crawlable` Bool,
    `has_subtitles` Bool,
    `is_ads_enabled` Bool,
    `is_comments_enabled` Bool,
    `description` String,
    `rich_metadata` Array(Tuple(call String, content String, subtitle String, title String, url String)),
    `super_titles` Array(Tuple(text String, url String)),
    `uploader_badges` String,
    `video_badges` String
)
ENGINE = ReplicatedMergeTree('/clickhouse/tables/{shard}/{database}/{table}', '{replica}')
ORDER BY (uploader, upload_date);

INSERT INTO replicated.youtube
SELECT * FROM default.youtube;
```

4_ Отдаем запросы как два файла

```
SELECT getMacro('replica'),
*
FROM remote('clickhouse1:9000,clickhouse2:9000,clickhouse3:9000', system.parts, 'default', '***')
FORMAT JSONEachRow;
```

```
{"getMacro('replica')":{"ch1","partition":"tuple()","name":"all_5_0","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Compact","active":1,"marks":"2","rows
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2229_1853","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"287
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2230_1854","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"287
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2235_1855","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"288
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2236_1856","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"288
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2237_1857","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"288
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2238_1858","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"288
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2239_1859","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"288
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2240_1860","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"288
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2241_1861","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"288
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2242_1862","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"289
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2243_1863","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"289
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2244_1864","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"289
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2245_1865","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"289
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2246_1866","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"289
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2247_1867","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"289
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2248_1868","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"289
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2249_1869","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"289
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2250_1870","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"290
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2251_1871","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"290
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2252_1872","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"290
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2253_1873","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"290
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2254_1874","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"290
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2255_1875","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"290
{"getMacro('replica')":{"ch1","partition":"202511","name":"202511_1_2256_1876","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":0,"marks":"290
```

```
SELECT *
FROM system.replicas
FORMAT JSONEachRow;
```

```
{
  "database": "replicated",
  "table": "youtube",
  "engine": "ReplicatedMergeTree",
  "is_leader": 1,
  "can_become_leader": 1,
  "is_readonly": 0,
  "readonly_start_time": null,
  "is_session_expired": 0,
  "future_parts": 0,
  "parts_to_check": 0,
  "zookeeper_name": "default",
  "zookeeper_path": "\\clickhouse\\tables\\01\\replicated\\youtube",
  "replica_name": "ch1",
  "replica_path": "\\clickhouse\\tables\\01\\replicated\\youtube\\replicas\\ch1",
  "columns_version": -1,
  "queue_size": 0,
  "inserts_in_queue": 0,
  "merges_in_queue": 0,
  "part_mutations_in_queue": 0,
  "queue_oldest_time": "1970-01-01 00:00:00",
  "inserts_oldest_time": "1970-01-01 00:00:00",
  "merges_oldest_time": "1970-01-01 00:00:00",
  "part_mutations_oldest_time": "1970-01-01 00:00:00",
  "oldest_part_to_get": "",
  "oldest_part_to_merge_to": "",
  "oldest_part_to_mutate_to": "",
  "log_max_index": "2",
  "log_pointer": "3",
  "last_queue_update": "2025-11-30 15:02:49",
  "absolute_delay": "0",
  "total_replicas": 3,
  "active_replicas": 3,
  "lost_part_count": "0",
  "last_queue_update_exception": "",
  "zookeeper_exception": "",
  "replica_is_active": {
    "ch3": 1,
    "ch1": 1,
    "ch2": 1
  }
}
```

5_Добавляем колонку

```
ALTER TABLE replicated.youtube ON CLUSTER replicated_cluster MODIFY TTL upload_date + INTERVAL 7 DAY;

SHOW CREATE TABLE replicated.youtube;
```

```
SHOW CREATE TABLE replicated.youtube
Query id: e1605178-55b7-435a-a9d9-dc56da0b073e

1. statement
CREATE TABLE replicated.youtube
(
  `id` String,
  `fetch_date` DateTime,
  `upload_date_str` String,
  `upload_date` Date,
  `title` String,
  `uploader_id` String,
  `uploader` String,
  `uploader_sub_count` Int64,
  `is_age_limit` Bool,
  `view_count` Int64,
  `like_count` Int64,
  `dislike_count` Int64,
  `is_crawlable` Bool,
  `has_subtitles` Bool,
  `is_ads_enabled` Bool,
  `is_comments_enabled` Bool,
  `description` String,
  `rich_metadata` Array(Tuple(
    call String,
    content String,
    subtitle String,
    title String,
    url String)),
  `super_titles` Array(Tuple(
    text String,
    url String)),
  `uploader_badges` String,
  `video_badges` String
)
ENGINE = ReplicatedMergeTree('/clickhouse/tables/{shard}/replicated/youtube', '{replica}')
ORDER BY (uploader, upload_date)
TTL upload_date + toIntervalDay(7)
SETTINGS index_granularity = 8192

1 row in set. Elapsed: 0.002 sec.
```